

Dalton Kutzen

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Education

Brigham Young University — B.S. Molecular Biology
Provo, Utah

Jan 2025 – Apr 2028

Experience

Bioinformatics Researcher

Aug 2026 – Present

[Terooatea Lab, Brigham Young University](#) · Provo, UT

- Developed computational methods and benchmark evaluations for high-dimensional single-cell RNA-seq (scRNA-seq) and pooled Perturb-seq screens (anchor-op).
- Implemented target-grouped nested cross-validation and matched signal-to-noise ratio (SNR) controls in Python to evaluate prediction performance against baseline models.
- Co-authored bioRxiv research on perturbation screen modeling and computational reproducibility.

Research Assistant

Sep 2025 – Present

[Grose Lab, Brigham Young University](#) · Provo, UT

- Prepared media, cultured bacteriophages with bacterial hosts, and conducted plaque and spot assays to evaluate phage activity and host interactions.
- Performed bioinformatic analyses to annotate and compare phage genomes, creating the **Phantom** Snakemake pipeline for reproducible annotation.
- Integrated wet lab microbiology techniques with computational genomics to investigate phage diversity, host range, and therapeutic applications.

IT Service Desk Supervisor

May 2026 – Present

[Brigham Young University — College of Life Sciences](#) · Provo, UT

- Lead frontline IT support for faculty, staff, and students across the College of Life Sciences, resolving high-complexity hardware, software, and systems incidents.
- Mentor student technicians in troubleshooting best practices and train new hires on Life Sciences IT infrastructure, tools, and clear user communication.
- Drove internal automation, documentation, and reporting initiatives that reduced repetitive workflows and standardized support delivery college-wide.

Projects

Self-Hosted Systems and Infrastructure

Dec 2025 - Present

- Built a Linux-based environment for code hosting, containers, services, and research project organization.
- Applied infrastructure-as-code and documentation practices to reduce configuration drift.

Skills

Area	Tools & Proficiencies
Computational Biology	Comparative genomics, phage genome annotation, Perturb-seq analysis, Snakemake, Conda, Biopython
Wet Lab & Molecular Biology	Phage culturing, plaque & spot assays, media preparation, agarose gel electrophoresis, host range testing
Programming & Data	Python, R, Bash, SQL, Git, pandas, tidyverse, ggplot2, Quarto
IT & Systems Infrastructure	IT Service Management (ITSM), Linux server administration, Proxmox, Docker, 1,000+ endpoint management, networking
Operations & Leadership	Good Manufacturing Practice (GMP), project management, technical mentoring, volunteer training (200+ individuals), CRM systems, documentation